Optimization for Deep Learning (OPT4DL)

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Min-max optimization and GANs

From Supervised Learning to Generative Modeling



- Recall so far we talked about training tasks of the form

$$\min f(w) = \mathbb{E}_{z\sim q}[\ell(w, z)] \tag{1}$$

  with the hope of finding a model that best fits the training data
- In supervised learning, the implication is that our parametric model $A(\cdot, w)$ learns the unknown rule between features and labels $y_i = A^*(x_i)$ such that $\|A^*(\cdot) - A(\cdot, w)\|$ is minimized
- In generative modeling the goal is to learn the data distribution p_z or learn the ability to sample from it.
- If we think of $q(\cdot)$ as the true rule, a natural solution then is to use a parametric distribution $p(\cdot, w)$ such that it approximates the true rule p_z well, i.e. $\|q(\cdot) - p(\cdot, w)\|$
- This is precisely the goal of variational inference:

$$\min_w f(w) = D(q\|p(w)) \tag{2}$$

  for some notion of distance or divergence between distributions, e.g. the KL divergence.

Distances among distributions



- f-divergence:

$$D_f(q\|p) = \mathbb{E}_p[f(\frac{q}{p})] = \int_z p(z) f(\frac{q(z)}{p(z)}) \cdot dz \tag{3}$$

  for f being a convex function. Examples include
  - KL divergence with $f(t) = t \log t$
  - TV distance with $f(t) = \frac{1}{2}|t-1|$
  - Jensen-Shannon Distance (JSD) a symmetrized and smoothed version KL: $D_{JSD}(q\|p) = \frac{1}{2} D_{KL}(p\|q) + \frac{1}{2} D_{KL}(q\|p)$

- a-Wasserstein distance or Kantorovich–Rubinstein metric:

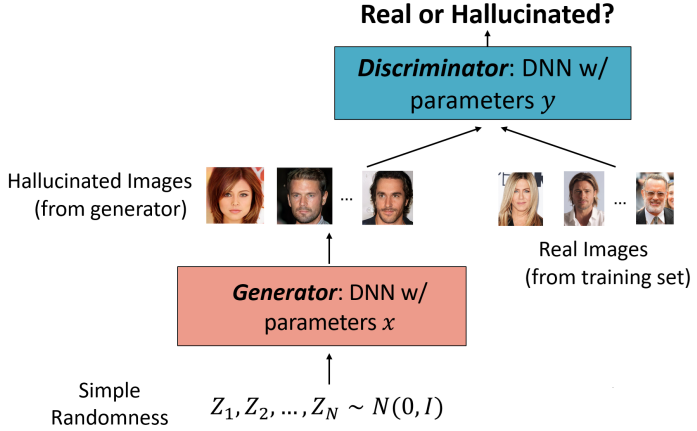
$$W_a(q,p) = \min_{\pi \in C(q,p)} \left(\mathbb{E}_{(z_1,z_2)\sim\pi} \|z_1 - z_2\|^a \right)^{\frac{1}{a}} \tag{4}$$

  for $a \geq 1$ where $C(q,p)$ is the set of all couplings (joint distributions) with marginals q and p.

- Turns out minimizing these is like solving a kind of game

Generative Adversarial Networks (GANs)

- p_w is good estimator of q if a skilled discriminator cannot distinguish $\tilde{z} \sim p_w$ from $z \sim q$
- This suggest building a zero-sum game between two players: one trying to generate fake data (the generator) and other trying to distinguish fake from real data (the discriminator)



GAN Game and Variational Inference



- Let D_y and G_x be the discriminator and generator with parameters y and x, respectively

$$\min_x \max_y U(x, y) := \mathbb{E}_{z\sim p}[\log D_y(z)] + \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[\log(1 - D_y(G_x(Z))] \tag{5}$$

- D_y is a binary classier and assigns high score ($\log D_y$) to real data and low score to fake data.
- G_x want to assign a high score to fake data by maximizing $\log(D_y(G_x(Z))$
- With simple calculation, we can show the best discriminator satisfies:

$$D_{y^*}(z) = \frac{q(z)}{q(z) + p_{G_x}(\cdot)} \tag{6}$$

- Plugging this into the problem above yields the optimal solution for the generator:

$$x^* = \operatorname*{argmin}_x 2D_{JSD}(q\|p_{G_x}) - \log 4 \tag{7}$$

- Thus in equilibrium $D_{y^*}(z) = 0.5$ and $p_{G_{x^*}}(z) = q(z)$
- So a natural question is can we go from a general f-divergence to a game?

f-GANs: Variational Representation of f-Divergences



- For any convex f define it's Fenchel dual $\tilde{f}$ (which is concave)

$$\tilde{f}(\tilde{t}) = \max_{t} \Big\{ \langle \tilde{t}, t \rangle - f(t) \Big\}, \qquad f(t) = \max_{\tilde{t}} \Big\{ \langle t, \tilde{t} \rangle - \tilde{f}(\tilde{t}) \Big\} \tag{8}$$

- We can use this to lower bound f-divergences:

$$\begin{aligned} D_f(q\|p) &= \int_z p(z) f\Big(\frac{q(z)}{p(z)}\Big) \cdot dz = \int_z p(z) \max_{\tilde{t}} \Big\{ \frac{q(z)}{p(z)} \tilde{t} - \tilde{f}(\tilde{t}) \Big\} \cdot dz \\ &\geq \max_{\tilde{t}} \Big\{ \int_z \Big(p(z) \frac{q(z)}{p(z)} \tilde{t} - \tilde{f}(\tilde{t}) \Big) \cdot dz \Big\} \\ &\geq \max_{\tilde{t} \in \mathcal{T}(z)} \Big\{ \int_z q(z) \tilde{t}(z) \cdot dz - \int_z p(z) \tilde{f}(\tilde{t}(z)) \cdot dz \Big\} \\ &= \max_{\tilde{t}(z) \in \mathcal{T}(z)} \mathbb{E}_{z \sim q}[\tilde{t}(z)] - \mathbb{E}_{z \sim p}[\tilde{f}(\tilde{t}(z))] \end{aligned} \tag{9}$$

where the first inequality is due to Jensen's inequality and convexity of max while the second inequality is by restricting the set of all $\tilde{t}$ to a class $\mathcal{T}$ of functions of z

f-GANs: Variational Representation of f-Divergences (Cont'd)



- Let us change $\tilde{t}(z)$ to $D_y(z)$ implying our function class is parametric with parameters y:

$$D_f(q\|p) \geq \max_y \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{z\sim p}[\tilde{f}(D_y(z))] \tag{10}$$

- Recall we think of q as data distribution and p as its estimator. If we assume p is parameterized via p_G for some function $G_x(Z)$ taking random Gaussian $Z \sim \mathcal{N}(0, I)$, fitting q by minimizing $D_f(q\|p)$ amounts to

$$\min_x \max_y \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[\tilde{f}(D_y(G_x(Z)))] \tag{11}$$

  generalizing GANs from JSD to arbitrary f-divergences.

- The bound is tight for $D_{y^*}(z) = f'(\frac{q(z)}{p(z)})$
- Generator is trying to minimize the divergence estimate, while the discriminator tries to tighten the lower bound.
- How about a-Wasserstein distance? Any benefit?

Wasserstein GAN



- Recall $D_f(q\|p) = \mathbb{E}_p[f(\frac{q}{p})]$ so the support of p has to cover the support of q. That is if $p(z_1) = 0$ we like $q(z_1) = 0$ too
- Otherwise discontinuity arises in f-divergences and they could become ∞ (e.g., KL).
- Turns out a-Wasserstein distances are nicer in such scenarios
- Using Kantorovich-Rubinstein duality, we represent this task as a minmax game. For $a = 1$

$$W_1(q,p) = \min_{\pi \in C(q,p)} \Big(\mathbb{E}_{(z_1,z_2)\sim\pi} \|z_1 - z_2\| \Big) = \max_{D:\|D\|_{Lip}\le 1} \mathbb{E}_{z\sim q}[D(z)] - \mathbb{E}_{z\sim p}[D(z)] \tag{12}$$

  where $\|D\|_{Lip} \le 1$ means D must have a Lipschitz constant at most 1
- This idea leads to Wasserstein GAN (WGAN)

$$\min_x \max_y \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[D_y(G_x(Z))] \tag{13}$$

- Lipschitzness is ensured by weight clipping or gradient penalty $\|\nabla_y D_y(z)\|$

Difficulties of Minmax Optimization

- Recall we showed for f-GAN training

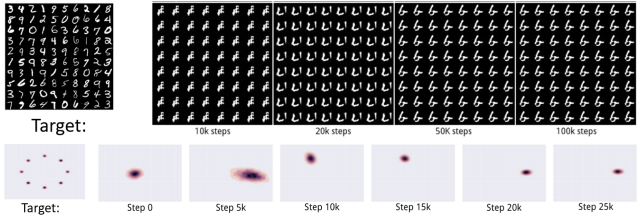
$$\min_x \max_y U(x,y) = \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[\tilde{f}(D_y(G_x(Z)))] \tag{14}$$

- And for WGAN training

$$\min_x \max_y U(x,y) = \mathbb{E}_{z\sim q}[D_y(z)] - \mathbb{E}_{Z\sim\mathcal{N}(0,I)}[D_y(G_x(Z))] \tag{15}$$

- Both are minmax optimization. One could use minmax version of (S)GD for training

$$x_{t+1} = x_t - \eta\nabla_x U(x_t, y_t), \qquad y_{t+1} = y_t + \eta\nabla_y U(x_t, y_t) \tag{16}$$



Minmax vs min



- The issue can be attributed to the fundamental difference between minmax and min problems.
- For minimization we interpreted GD a simple forward Euler discretization of gradient flow

$$w_{t+1} = w_t - \eta \nabla f(w_t), \qquad \dot{w} = -\nabla f(w) \tag{17}$$

- Let $v = [x, y]^\top$ and define $F(v) = [\nabla_x U(x_t, y_t), -\nabla_y U(x_t, y_t)]^\top$. Minmax GD becomes

$$v_{t+1} = v_t - \eta F(v_t), \qquad \dot{v} = -F(v) \tag{18}$$

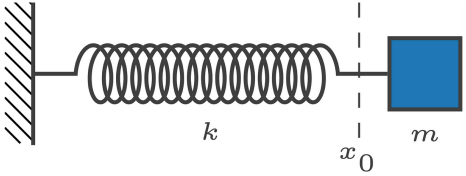
- The look very similar, but there is a fundamental difference! ∇f is the gradient of a scalar function but F is not
- Jacobian of ∇f, $J_{\nabla f}$ is symmetric (the Hessian $\nabla^2 f$) while the Jacobian of F, J_F is not
- The Jacobian controls the local dynamics: how the flow $F(v)$ curves, rotates, contracts, or expands locally.

Dissipative vs. Conservative Systems

- Note $J_F = \frac{J_F + J_F^\top}{2} + \frac{J_F - J_F^\top}{2}$, i.e., sum of symmetric and skew-symmetric matrices
- when the flow is $\dot{w} = -\nabla f(w)$ there is no skew-symmetric component and under mild condition the system is dissipating energy.
- When the flow is $\dot{v} = -F(v)$, the system might conserve energy and the skew-symmetric component determines the rotation.
- The most simple example is harmonic oscillator

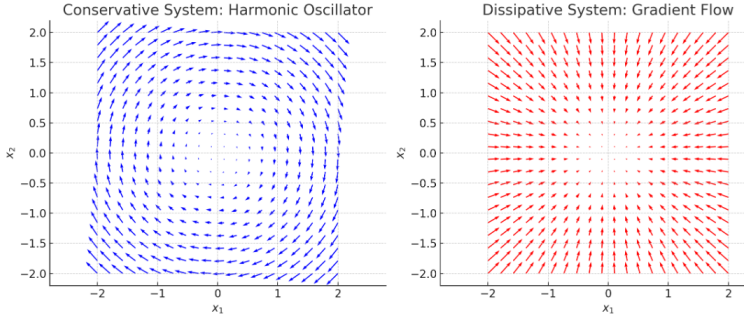
$$\ddot{w} = -w \qquad \equiv \qquad \dot{x} = y, \quad \dot{y} = -x \qquad \equiv \qquad \dot{v} = -F(v) \tag{19}$$

for which the Jacobian is purely skew-symmetric. It corresponds to the simplest minmax problem: $\min_x \max_y \; U(x, y) = xy$



Dissipative vs. Conservative Systems (Cont'd)

- Following the flow might not help in skew-symmetric dynamics
- Simple Euler forward discretization (GD in optimization) converges nicely to a stable point for dissipative systems while does not work well for conservative systems as it pushes us around (and potentially away) the stable solution
- Thus, for minmax problems we generally needs other classes of method



Negative Momentum and Predictability

- Recall we talked a bout momentum for min problems. Intuitively, we assume usefulness of gradient direction and we double down by introducing positive momentum
- For minmax problems, the gradient direction has a mistake and we can correct that by introducing negative momentum.
- This idea leads to the update $v_{t+1} = v_t - \eta F(v_{t+1})$
- This is precisely the Euler Backward discretization of the flow and it is generally implicit.
- Also called proximal point method since if $F \to \nabla f$ for some convex function f

$$w_{t+1} = w_t - \eta \nabla f(w_{t+1}) \qquad \equiv \qquad w_{t+1} = \arg\min_w f(w) + \frac{1}{2\eta}\|w - w_t\|^2 \tag{20}$$



Mid-Point Discretization and Extra Gradient



- Consider the flow

$$\dot{v} = -F(v) \qquad \equiv \qquad v_{t+1} = v_t - \int_t^{t+1} F(v_\tau) \cdot d\tau \tag{21}$$

- Forward Euler (GD) approximates

$$v_{t+1} = v_t - \int_t^{t+1} F(v_\tau) \cdot d\tau \approx \int_t^{t+1} F(v_t) \cdot d\tau = v_t - \eta F(v_t) \tag{22}$$

- Backward Euler (Proximal Point) approximates

$$v_{t+1} = v_t - \int_t^{t+1} F(v_\tau) \cdot d\tau \approx \int_t^{t+1} F(v_{t+1}) \cdot d\tau = v_t - \eta F(v_{t+1}) \tag{23}$$

Mid-Point Discretization and Extra Gradient (Cont'd)

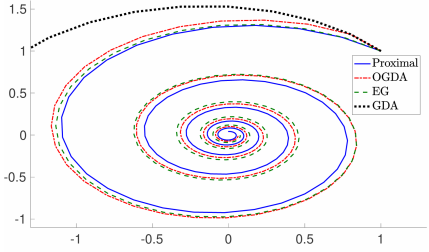
- More generally for any $u \in (t, t+1)$ and in particular for $u = 1/2$ we can do

$$v_{t+1} = v_t - \int_t^{t+1} F(v_\tau) \cdot d\tau \approx \int_t^{t+1} F(v_{t+\frac{1}{2}}) \cdot d\tau = v_t - \eta F(v_{t+\frac{1}{2}}) \tag{24}$$

  and find $v_{t+\frac{1}{2}}$ via Forward Euler to obtain an explicit update

$$v_{t+1} = v_t - \eta F(v_{t+\frac{1}{2}}), \qquad v_{t+\frac{1}{2}} = v_t - \eta F(v_t) \tag{25}$$

- Called Extra Gradient (EG), approximates PP by doing $F(v_{t+1}) \approx F(v_t - \eta F(v_t))$



Optimistic Gradient Descent



- Consider a field which is Lipschitz, meaning the variations are controlled: $F(v_{t+1}) \approx F(v_t)$.
- This yields GD. We hope to get a tighter approximation by doubling down on our optimism towards controlled variation of F
- $F(v_{t+1}) \approx F(v_t)$ is effectively the zeroth-order Taylor approximation. Consider the first-order Taylor approximation instead

$$F(v_{t+1}) \approx F(v_t) + J_F(v_t)(v_{t+1} - v_t), \qquad J_F(v_t) \approx \frac{F(v_t) - F(v_{t-1})}{\|v_t - v_{t-1}\|} \cdot e^\top \tag{26}$$

  where $e = \frac{v_t - v_{t-1}}{\|v_t - v_{t-1}\|}$ is the unit vector along the direction of $v_t - v_{t-1}$
- Let us be optimistic one last time and assume $v_{t+1} - v_t \approx v_t - v_{t-1}$, leading to

$$F(v_{t+1}) \approx F(v_t) + \frac{F(v_t) - F(v_{t-1})}{\|v_t - v_{t-1}\|} \frac{(v_t - v_{t-1})^\top}{\|v_t - v_{t-1}\|}(v_t - v_{t-1}) = F(v_t) + F(v_t) - F(v_{t-1}) \tag{27}$$

- This serves as the update vector in Optimistic GD: $v_{t+1} = v_t - 2\eta F(v_t) + \eta F(v_{t-1})$